MISSING DATA & STATISTICAL INFERENCE

A Simulation-Based Study of MCAR, MAR and MNAR

Comparing Seven Missing-Data Handling Methods Within Each Mechanism

Project Report

M.Sc. Statistics

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Abstract

This project investigates how missing-data handling methods affect regression estimation and statistical inference when observations are incomplete. A complete dataset of $n = 100$ observations was simulated from a known regression model,

$$Y = 5 + 2X_1 - 1.5X_2 + X_3 + \epsilon,$$

so that the true coefficients were known. Missing values were then deliberately introduced in the response variable Y under three missingness mechanisms: Missing Completely At Random (MCAR), Missing At Random (MAR), and Missing Not At Random (MNAR).

For each mechanism, the same incomplete dataset was given to seven methods: complete-case analysis, mean imputation, hot-deck imputation, regression imputation, stochastic regression imputation, Multiple Imputation by Chained Equations (MICE) using predictive mean matching (PMM), and the Expectation-Maximization (EM) algorithm. The methods were compared separately for MCAR, MAR and MNAR.

The regression model estimates all four coefficients, but the detailed inferential analysis focuses on β_1, the coefficient of X_1, whose true value is 2. Performance is examined using the estimated coefficient, bias, bootstrap standard error, and 95% confidence interval. MICE additionally combines uncertainty across five completed datasets using Rubin's rules.

The clearest pattern in the simulation is the large downward distortion produced by mean imputation. Under MCAR, for example, mean imputation produced $\hat{\beta}_1 = 1.089$, compared with the true value of 2, whereas complete-case analysis produced 1.785. Under MAR, MICE produced a pooled estimate of 1.797, while mean imputation produced 1.307. Under MNAR, the estimates were also below the true value in the simulated dataset, with the largest deviation again occurring with mean imputation. The results show how the choice of missing-data treatment can change both the estimated relationship and the associated uncertainty.

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1. Introduction

Missing data are a common feature of real statistical datasets. Once observations are incomplete, the analysis may change not only because fewer observations are available, but also because the way missing observations are handled can alter the relationships among variables and the uncertainty of the resulting estimates.

A central distinction in missing-data analysis is the mechanism that generated the missingness. The standard classification is MCAR, MAR, and MNAR. Under MCAR, missingness does not depend on the data. Under MAR, missingness may depend on information that remains observed. Under MNAR, missingness may also depend on the value that is itself missing. The course material presents these mechanisms as a nested hierarchy,

$$\text{MCAR} \subset \text{MAR} \subset \text{MNAR},$$

with MCAR the most restrictive condition and MNAR the most general.

The practical issue is that there is no single way to handle missing observations. A researcher might delete incomplete cases, replace missing values with a mean, borrow values from donors, predict them from a regression model, create multiple imputations, or estimate model parameters directly using likelihood-based methods. These choices can lead to different estimates and different inferential conclusions.

In this study, MCAR, MAR and MNAR define three settings for the experiment. Within each setting, the seven missing-data handling methods are applied to the same incomplete dataset and their regression results are compared.

2. Objective and Research Question

2.1 Objective

The objective is to examine how different missing-data handling methods affect regression estimation and statistical inference when applied to the same incomplete dataset under a specified missingness mechanism.

The study evaluates both the location of the estimate and the uncertainty around it. In particular, the analysis examines:

$$\hat{\beta}_1, \qquad \text{Bias}(\hat{\beta}_1), \qquad SE(\hat{\beta}_1), \qquad 95\%\, CI.$$

Although the fitted regression contains four coefficients, β_1 is selected as the primary parameter of interest for the detailed inference comparison. Its true value is known to be

$$\boxed{\beta_1 = 2}.$$

2.2 Research Question

How does the missing-data handling method affect the estimated regression relationship and the associated statistical uncertainty when the missingness mechanism

is held fixed?

This question is examined three times: once for MCAR, once for MAR, and once for MNAR.

3. Data-Generating Model

To create a controlled experiment, a complete dataset was generated before any values were made missing. The response was generated from the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \epsilon_i.$$

The parameters used in the simulation were

$$\beta_0 = 5, \qquad \beta_1 = 2, \qquad \beta_2 = -1.5, \qquad \beta_3 = 1.$$

The predictors and random error were generated as

$$X_1, X_2, X_3 \overset{ind}{\sim} N(0,1), \qquad \epsilon \sim N(0,1).$$

A sample of $n = 100$ observations was generated. The complete dataset therefore represents the hypothetical data that would have been available if no information were missing.

The benefit of simulation is that the true parameter is known. With real data, the true value of a regression coefficient is not available as a benchmark. Here, however, an estimate such as $\hat{\beta}_1 = 1.70$ can be directly compared with the known truth $\beta_1 = 2$.

3.1 Baseline Complete-Data Regression

Before introducing missingness, the regression was fitted to the complete dataset. The resulting coefficients were:

Table 1: Baseline complete-data regression.

Parameter	True value	Complete-data estimate
β_0	5.000	5.0875
β_1	2.000	1.8227
β_2	-1.500	-1.5382
β_3	1.000	1.0269

The fact that the complete-data estimate of β_1 is 1.8227 rather than exactly 2 is expected. The simulated dataset is a sample, so sampling variation exists even before missingness is introduced. This is important when interpreting later deviations from the true value: not all deviation from 2 is necessarily caused by missing data.

4. Missing-Data Mechanisms

After generating the complete dataset, missingness was introduced only in Y. The same complete dataset was used as the starting point for all three mechanisms. This means that, within each mechanism, all seven methods receive the same incomplete data.

4.1 MCAR

Under MCAR, every observation has the same probability of having its Y value made missing. The simulation used

$$P(R_i = 0) = 0.20,$$

where $R_i = 0$ denotes a missing response.

For each of the 100 observations, a random number between 0 and 1 was generated. Whenever the number was below 0.20, the corresponding Y value was replaced by 'NaN'.

In this particular realization, 25 observations had missing Y, so 75 observations retained an observed response. The realized percentage was therefore 25%, even though the specified probability was 20%. A probability of 20% does not guarantee exactly 20 missing observations in a random sample of 100.

4.2 MAR

Under MAR, missingness depends on observed information. The simulation made the missingness probability depend on X_1:

$$P(R_i = 0 \mid X_{1i}) = \begin{cases} 0.30, & X_{1i} < 0, \\ 0.10, & X_{1i} \geq 0. \end{cases}$$

Thus, observations with negative X_1 values were more likely to lose their Y value than observations with non-negative X_1.

In this realization, 25 observations had missing Y. A useful diagnostic confirmed the intended relationship: the mean X_1 among observations with observed Y was approximately 0.055, while the mean X_1 among observations with missing Y was approximately -0.581.

4.3 MNAR

Under MNAR, missingness depends directly on the value that becomes missing. The simulation first standardized the original response:

$$Y_i^* = \frac{Y_i - \bar{Y}}{s_Y}.$$

The probability of missingness was then defined as

$$p_i = \operatorname{expit}\left(-1.3863 + 0.5Y_i^*\right),$$

where

$$\text{expit}(z) = \frac{1}{1+e^{-z}}.$$

Because Y_i^* appears directly in the probability formula, observations with larger original Y values have a higher probability of becoming missing. The probabilities are then compared with a new random number for each observation.

In the particular realization, 16 observations had missing Y. Using the original complete dataset, the mean Y among observations that became missing was approximately 5.157, while the mean among observations that remained observed was approximately 4.888. This is consistent with the direction built into the MNAR mechanism.

Table 2: Missingness mechanisms and realized sample sizes.

Mechanism	Dependence	Missing Y	Observed Y
MCAR	None	25	75
MAR	Observed X_1	25	75
MNAR	Original Y	16	84

The key distinction is therefore not the percentage of missing observations. The mechanism is defined by *what the probability of missingness depends on.*

5. Missing-Data Handling Methods

For each missingness mechanism, the same seven approaches were applied.

5.1 Complete-Case Analysis

Rows with missing Y were removed and the regression was fitted only to the remaining observations. No artificial values were created, but information was discarded.

Under MCAR, complete-case observations form a random subset of the complete data in the usual sense. Under MAR or MNAR, the validity of complete-case analysis requires stronger assumptions; in general, the course material emphasizes MCAR for the usual unbiasedness result.

5.2 Mean Imputation

Each missing response was replaced by the mean of the observed responses:

$$Y_i^{imp} = \bar{Y}_{obs} \quad \text{for missing } Y_i.$$

This is simple, but all missing observations receive the same value. The course material shows that under MCAR this reduces the variance of the imputed outcome and attenuates correlations with predictors. Consequently, mean imputation can weaken regression relationships.

5.3 Hot-Deck Imputation

For each missing observation, an observed donor with similar values of (X_1, X_2, X_3) was found using nearest-neighbour squared distance,

$$D_{ij} = (X_{1i} - X_{1j})^2 + (X_{2i} - X_{2j})^2 + (X_{3i} - X_{3j})^2.$$

The Y value of the closest observed donor was then copied to the missing observation.

Because donor values are actual observed outcomes, hot-deck does not replace every missing value with the same constant. However, it remains a single-imputation method and therefore does not fully represent imputation uncertainty.

5.4 Regression Imputation

A regression model was fitted to observations with observed Y values:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon.$$

For a missing response, the fitted model supplied

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i}.$$

The predicted value was then used as the imputed outcome.

5.5 Stochastic Regression Imputation

Stochastic regression imputation added residual variation to the regression prediction:

$$Y_i^{imp} = \hat{Y}_i + e_i^*,$$

where e_i^* was sampled from observed regression residuals. This allows the imputed observations to retain random outcome variation rather than lying exactly on the fitted regression relationship.

5.6 MICE Using Predictive Mean Matching

MICE normally handles several incomplete variables by cycling through them one at a time, fitting a suitable conditional model at each step. In this project only Y is incomplete, so the implementation is a simplified PMM-based multiple-imputation procedure.

The project used

$$m = 5$$

completed datasets and

$$k = 5$$

nearest potential donors in predictive mean matching. Because only Y is incomplete and the predictors are fully observed, this is a simplified one-variable MICE/PMM setting.

The initial missing values were filled with a temporary mean. A regression of observed Y on the fully observed predictors was then fitted. For each missing observation, the predicted value was

compared with predicted values of observed cases; the five closest observed cases were treated as possible donors, and one donor's actual observed Y was randomly selected. This process generated one completed dataset. The process was repeated to generate five completed datasets.

Each completed dataset was analyzed separately. The five resulting $\hat{\beta}_1$ values were then pooled using Rubin's rules. If W denotes the average within-imputation variance and B the between-imputation variance, the total variance is

$$T = W + \left(1 + \frac{1}{m}\right) B.$$

The pooled standard error is

$$SE_{pooled} = \sqrt{T}.$$

5.7 Expectation-Maximization

EM estimates the parameters directly from the incomplete data. The E-step calculates expected quantities involving the missing responses using current parameter estimates, and the M-step updates the regression coefficients and error variance.

The algorithm was iterated until changes in the parameters and variance were sufficiently small. In this project the final EM results were followed by bootstrap inference for β_1.

6. Evaluation Framework

6.1 Bias

The bias in a single simulation is measured relative to the known truth:

$$\text{Bias}(\hat{\beta}_1) = \hat{\beta}_1 - \beta_1 = \hat{\beta}_1 - 2.$$

Absolute bias is

$$|\hat{\beta}_1 - 2|.$$

A negative bias means that the estimate is below 2.

6.2 Bootstrap Standard Error

For the bootstrap-based methods, 500 bootstrap samples were generated by resampling observations with replacement within the corresponding method-specific workflow. The analysis was repeated for each bootstrap sample, giving

$$\hat{\beta}_1^{(1)}, \hat{\beta}_1^{(2)}, \ldots, \hat{\beta}_1^{(500)}.$$

The bootstrap standard error was estimated as

$$SE_{boot} = SD\left(\hat{\beta}_1^{(1)}, \ldots, \hat{\beta}_1^{(500)}\right).$$

6.3 95% Confidence Intervals

The bootstrap confidence intervals use the percentile method:

$$CI_{95\%} = (q_{0.025}, q_{0.975}),$$

where q_p denotes the pth percentile of the bootstrap distribution.

For MICE, the interval is obtained from the pooled estimate and Rubin's rules rather than treating the five imputations as one ordinary bootstrap sample.

7. Results

The results are presented separately for MCAR, MAR and MNAR. Within each mechanism, the seven methods use the same incomplete dataset.

7.1 MCAR: Compare All Seven Methods

The MCAR dataset contained 25 missing Y values and 75 observed responses. The results for the focal coefficient are shown below.

Table 3: MCAR: comparison of all seven methods.

Method	$\hat{\beta}_1$	SE	CI lower	CI upper	Bias
Complete Case	1.785	0.136	1.527	2.025	-0.215
Mean Imputation	1.089	0.229	0.649	1.565	-0.911
Hot-Deck	1.658	0.165	1.262	1.918	-0.342
Regression Imputation	1.785	0.136	1.527	2.025	-0.215
Stochastic Regression	1.706	0.140	1.487	2.046	-0.294
MICE	1.659	0.114	1.433	1.885	-0.341
EM	1.785	0.136	1.527	2.025	-0.215

Mean imputation gives $\hat{\beta}_1 = 1.089$, about 0.91 below the true value. Its confidence interval also lies entirely below 2.

Complete-case analysis gives 1.785, only about 0.038 below the complete-data estimate of 1.823. Regression imputation and EM give essentially the same coefficient in this setup. Hot-deck and MICE both give estimates around 1.66, while stochastic regression gives 1.706.

The intervals also differ across methods. The complete-case, regression-imputation, stochastic-regression and EM intervals extend to 2 or above, while the intervals for mean imputation, hot-deck and MICE end below 2.

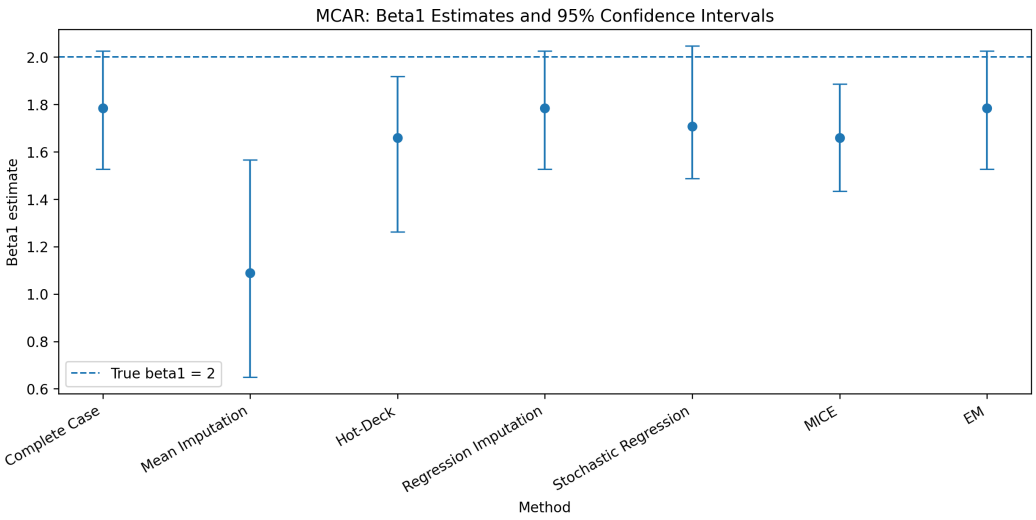


Figure 1: MCAR: β_1 estimates and 95% confidence intervals by method.

7.2 MAR: Compare All Seven Methods

The MAR dataset also contained 25 missing Y values. Here, missingness was related to the observed predictor X_1.

Table 4: MAR: comparison of all seven methods.

Method	$\hat{\beta}_1$	SE	CI lower	CI upper	Bias
Complete Case	1.777	0.136	1.532	2.055	-0.223
Mean Imputation	1.307	0.186	0.915	1.659	-0.693
Hot-Deck	1.787	0.148	1.416	1.996	-0.213
Regression Imputation	1.777	0.136	1.532	2.055	-0.223
Stochastic Regression	1.759	0.147	1.499	2.065	-0.241
MICE	1.797	0.094	1.612	1.982	-0.203
EM	1.777	0.136	1.532	2.055	-0.223

Mean imputation gives a substantially lower estimate of 1.307, with bias -0.693. Its interval, $(0.915, 1.659)$, lies entirely below 2.

The remaining estimates are much closer together. MICE gives a pooled estimate of 1.797 with absolute bias 0.203. Hot-deck gives 1.787, stochastic regression 1.759, and complete case, regression imputation and EM all give approximately 1.777.

For MICE, the five imputed datasets produced

$$1.7683, \quad 1.8304, \quad 1.7944, \quad 1.7991, \quad 1.7925,$$

which pool to

$$\hat{\beta}_1 = 1.7969.$$

The within-imputation variance was 0.008294 and the between-imputation variance was 0.000493, producing a pooled SE of 0.0943. The 95% interval was $(1.612, 1.982)$.

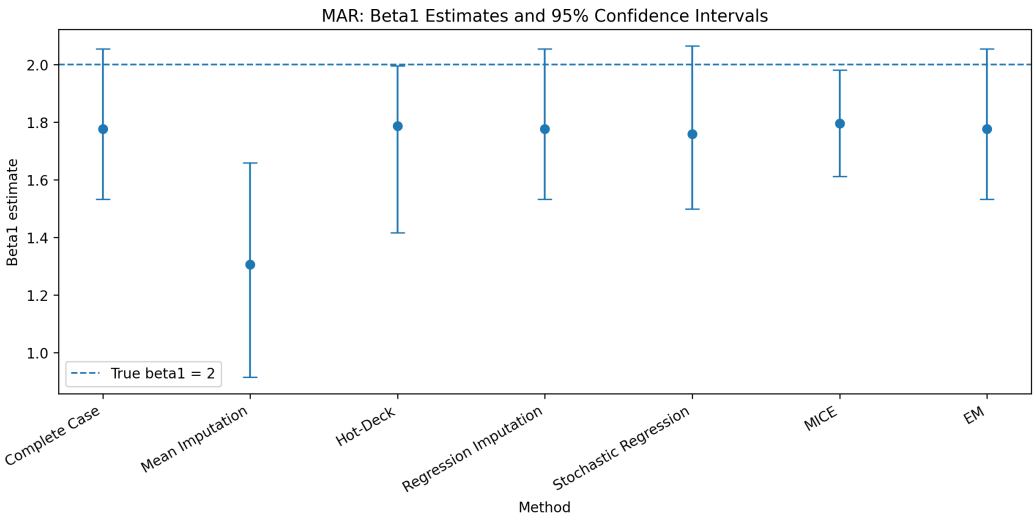


Figure 2: MAR: β_1 estimates and 95% confidence intervals by method.

7.3 MNAR: Compare All Seven Methods

The MNAR dataset contained 16 missing Y values. Missingness was directly related to the original response value.

Table 5: MNAR: comparison of all seven methods.

Method	$\hat{\beta}_1$	SE	CI lower	CI upper	Bias
Complete Case	1.749	0.122	1.501	1.981	-0.251
Mean Imputation	1.413	0.170	1.047	1.702	-0.587
Hot-Deck	1.722	0.123	1.469	1.954	-0.278
Regression Imputation	1.749	0.122	1.501	1.981	-0.251
Stochastic Regression	1.719	0.124	1.517	1.987	-0.281
MICE	1.722	0.108	1.510	1.933	-0.278
EM	1.749	0.122	1.501	1.981	-0.251

Mean imputation gives $\hat{\beta}_1 = 1.413$, with bias -0.587. The other six methods lie between about 1.719 and 1.749.

The MICE results were

$$1.7640, \quad 1.6876, \quad 1.7478, \quad 1.7083, \quad 1.7004,$$

with pooled estimate

$$\hat{\beta}_1 = 1.7216.$$

The within-imputation variance was 0.010295, the between-imputation variance was 0.001067, and the pooled SE was 0.1076. The resulting interval was $(1.510, 1.933)$.

All seven MNAR intervals lie below 2 in the simulated dataset.

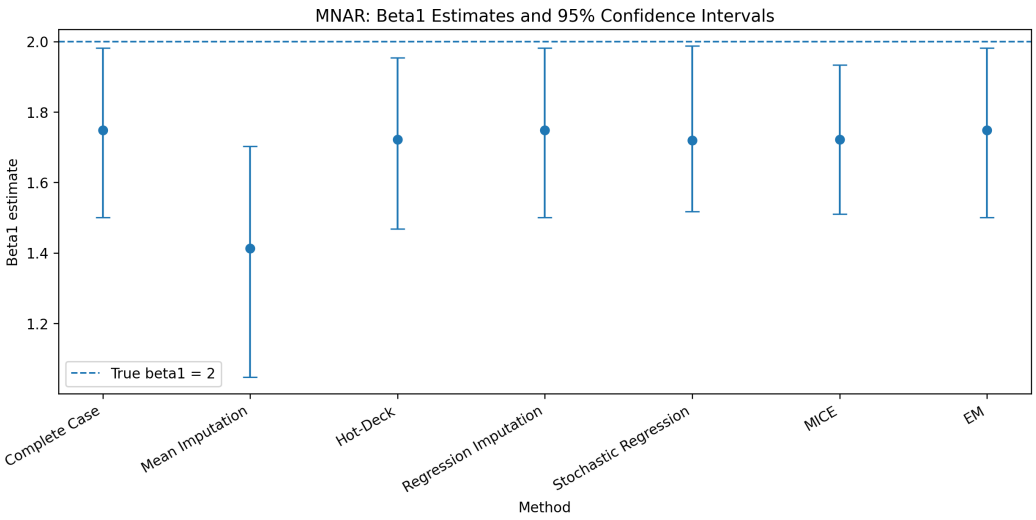


Figure 3: MNAR: β_1 estimates and 95% confidence intervals by method.

7.4 Cross-Method Summary

The main conclusions are obtained by comparing methods within each mechanism.

Table 6: Absolute bias of β_1 within each missingness mechanism.

Method	MCAR	MAR	MNAR
Complete Case	0.215	0.223	0.251
Mean Imputation	0.911	0.693	0.587
Hot-Deck	0.342	0.213	0.278
Regression Imputation	0.215	0.223	0.251
Stochastic Regression	0.294	0.241	0.281
MICE	0.341	0.203	0.278
EM	0.215	0.223	0.251

This table should be read column by column. Within MCAR, for example, all seven entries describe methods applied to the same MCAR dataset. The same interpretation applies to MAR and MNAR.

The dominant pattern is the larger absolute bias associated with mean imputation. The other methods are more tightly clustered in all three settings, although their estimates and uncertainty differ.

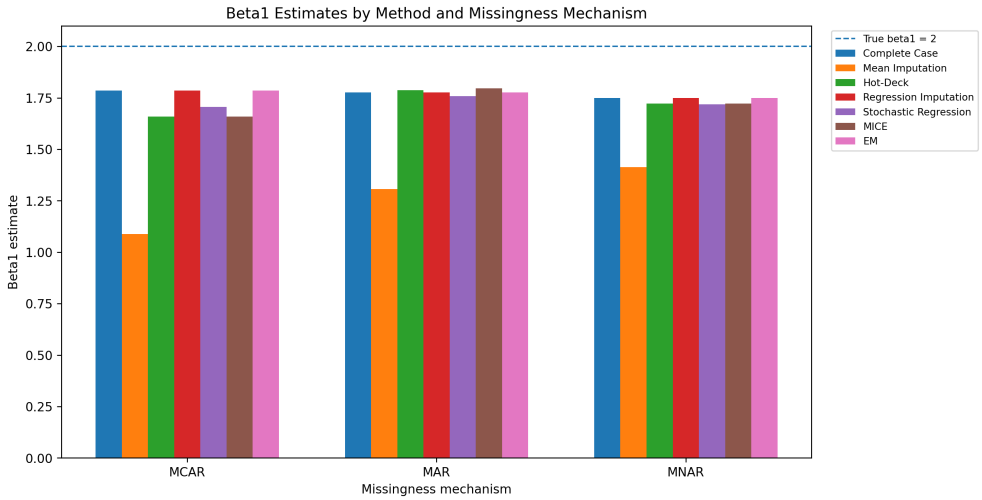


Figure 4: Overview of β_1 estimates, with MCAR, MAR and MNAR shown on the x-axis and the seven methods grouped within each mechanism.

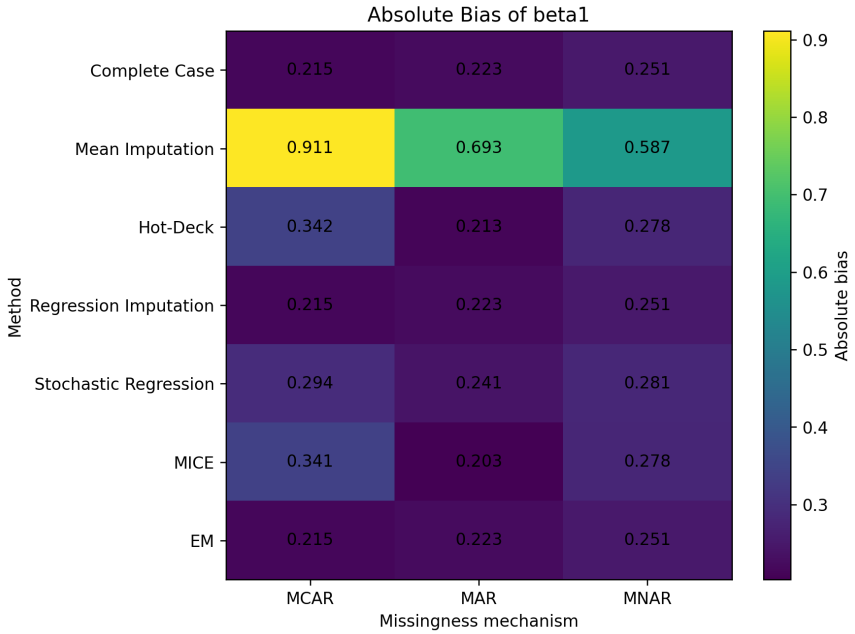


Figure 5: Absolute bias of β_1. Lower values indicate estimates closer to the known true value of 2 in this particular simulation.

8. Discussion

8.1 Mean Imputation Stands Out

Across MCAR, MAR and MNAR, mean imputation is the method that most clearly changes the regression result. It replaces all missing responses with a common value, removing variation from the imputed part of the response distribution. In the simulation this produces a strong downward movement in $\hat{\beta}_1$.

Under MCAR, the estimate drops to 1.089; under MAR it is 1.307; under MNAR it is 1.413.

The absolute biases are therefore 0.911, 0.693 and 0.587, respectively. These are substantially larger than the corresponding biases for the other methods in the same datasets.

The result is consistent with the theoretical course material, which shows that mean imputation shrinks outcome variance and attenuates correlations under MCAR. In a regression setting, reducing the natural variation in Y can weaken the apparent relationship between Y and its predictors.

8.2 Complete Case, Regression Imputation and EM

In this setup, complete-case analysis, regression imputation and EM produce the same point estimates. The procedures are different, but only Y is incomplete and all predictors are fully observed, which leads to the same fitted coefficient values here.

Complete-case analysis simply discards the missing-response rows. Regression imputation predicts their Y values from the observed predictors. EM works through expected quantities for the missing outcomes while iteratively updating the model. Despite these conceptual differences, their fitted coefficient estimates coincide in this simulation.

This is an important example of why numerical equality does not necessarily imply methodological equality.

8.3 Hot-Deck and Stochastic Regression

Hot-deck retains actual observed Y values by borrowing values from similar donors. Stochastic regression also preserves additional outcome variability through sampled residuals. Both therefore behave differently from deterministic mean replacement.

The MCAR hot-deck estimate was 1.658 and the stochastic-regression estimate was 1.706. Under MAR they were 1.787 and 1.759, respectively. Under MNAR they were 1.722 and 1.719. The estimates are reasonably close to one another in several cases, but their confidence intervals and exact sampling behavior differ.

8.4 MICE and Imputation Uncertainty

MICE represents the missing responses through several completed datasets rather than a single imputed dataset. Five completed datasets were generated, producing five values of $\hat{\beta}_1$.

For MAR, these values were tightly clustered between 1.768 and 1.830. The between-imputation variance was only 0.000493, so the five plausible completed datasets led to similar regression estimates. For MCAR, the five values ranged from 1.608 to 1.724, giving a larger between-imputation variance of 0.001923. For MNAR, the range was approximately 1.688 to 1.764, with between-imputation variance 0.001067.

Rubin's rules then combine within-imputation and between-imputation uncertainty. The pooled variance therefore includes both the usual regression variance within each completed dataset and the variation across imputations.

8.5 Why MNAR Needs Extra Caution

In the MNAR setting, the probability of missingness depends on the original Y value itself. The observed data alone do not determine this missingness process, so additional assumptions are needed to identify an MNAR model.

The estimates obtained here, including the MICE estimate of 1.722, should therefore be interpreted in the context of the specified simulation mechanism rather than as evidence that a standard MAR-based procedure resolves MNAR.

8.6 Point Estimate Versus Inference

The confidence intervals show that the point estimate and its uncertainty provide different information. Under MAR, MICE gives $\hat{\beta}_1 = 1.797$ with interval $(1.612, 1.982)$, whereas stochastic regression gives 1.759 with interval $(1.499, 2.065)$. The point estimates are similar, but the resulting intervals are not.

9. Inference Summary

The central inferential quantities are summarized below.

Table 7: Inference summary for β_1. CI = 95% confidence interval.

Mechanism	Method	$\hat{\beta}_1$	SE	95% CI
MCAR	Complete Case	1.785	0.136	(1.527, 2.025)
MCAR	Mean	1.089	0.229	(0.649, 1.565)
MCAR	Hot-Deck	1.658	0.165	(1.262, 1.918)
MCAR	Regression	1.785	0.136	(1.527, 2.025)
MCAR	Stochastic Regression	1.706	0.140	(1.487, 2.046)
MCAR	MICE	1.659	0.114	(1.433, 1.885)
MCAR	EM	1.785	0.136	(1.527, 2.025)
MAR	Complete Case	1.777	0.136	(1.532, 2.055)
MAR	Mean	1.307	0.186	(0.915, 1.659)
MAR	Hot-Deck	1.787	0.148	(1.416, 1.996)
MAR	Regression	1.777	0.136	(1.532, 2.055)
MAR	Stochastic Regression	1.759	0.147	(1.499, 2.065)
MAR	MICE	1.797	0.094	(1.612, 1.982)
MAR	EM	1.777	0.136	(1.532, 2.055)
MNAR	Complete Case	1.749	0.122	(1.501, 1.981)
MNAR	Mean	1.413	0.170	(1.047, 1.702)
MNAR	Hot-Deck	1.722	0.123	(1.469, 1.954)
MNAR	Regression	1.749	0.122	(1.501, 1.981)
MNAR	Stochastic Regression	1.719	0.124	(1.517, 1.987)
MNAR	MICE	1.722	0.108	(1.510, 1.933)
MNAR	EM	1.749	0.122	(1.501, 1.981)

10. Limitations

The numerical results are conditional on the simulated sample and the missingness patterns generated for the three settings. A different random sample could produce different coefficient estimates.

The data-generating model is also deliberately simple: the predictors and errors are normally distributed and the relationship is linear. More complex data could lead to different behaviour.

Only the response variable Y is incomplete, so the MICE procedure is a simplified one-variable PMM application rather than a general multiple-variable chained-equations problem.

Finally, the MNAR mechanism is specified directly in the simulation. With real data, an MNAR process cannot generally be determined from the observed data alone; additional modeling assumptions or sensitivity analysis would be required.

11. Conclusion

The study compared seven missing-data handling approaches separately under MCAR, MAR and MNAR. The results show that the handling method can materially change both the estimated regression coefficient and its uncertainty.

Mean imputation produced the largest downward shift in β_1 in all three settings. The remaining methods produced estimates that were closer to the complete-data relationship in these simulated samples, although their confidence intervals differed.

MICE incorporated variation across multiple completed datasets through Rubin's rules, while EM provided a likelihood-based alternative. In the present one-incomplete-variable setting, EM produced the same coefficient estimates as complete-case and regression-imputation analyses.

The MNAR setting also illustrates the role of assumptions in missing-data analysis: when missingness depends on the unobserved response, additional information or assumptions are needed to determine the missingness process.

A. Additional MICE Details

The MICE results are particularly useful for understanding how multiple imputation works numerically.

Table 8: Five MICE β_1 estimates and pooled results.

Mechanism	m_1	m_2	m_3	m_4	m_5	Pooled	W	B
MCAR	1.679	1.645	1.608	1.641	1.724	1.659	0.010727	0.001923
MAR	1.768	1.830	1.794	1.799	1.792	1.797	0.008294	0.000493
MNAR	1.764	1.688	1.748	1.708	1.700	1.722	0.010295	0.001067

For MCAR,

$$W = 0.010727, \qquad B = 0.001923, \qquad SE_{pooled} = 0.114170.$$

For MAR,

$$W = 0.008294, \qquad B = 0.000493, \qquad SE_{pooled} = 0.094262.$$

For MNAR,

$$W = 0.010295, \qquad B = 0.001067, \qquad SE_{pooled} = 0.107590.$$

These numbers make the role of multiple imputation concrete. The within-imputation variance reflects ordinary regression uncertainty inside each completed dataset, while the between-imputation variance reflects the fact that different plausible imputations do not produce exactly the same regression coefficient.

B. EM Convergence Summary

The EM algorithm converged to the following β_1 estimates in the implementation:

Table 9: EM convergence summary.

Mechanism	Iterations to convergence	Final β_1
MCAR	20	1.784723
MAR	20	1.776896
MNAR	14	1.748686

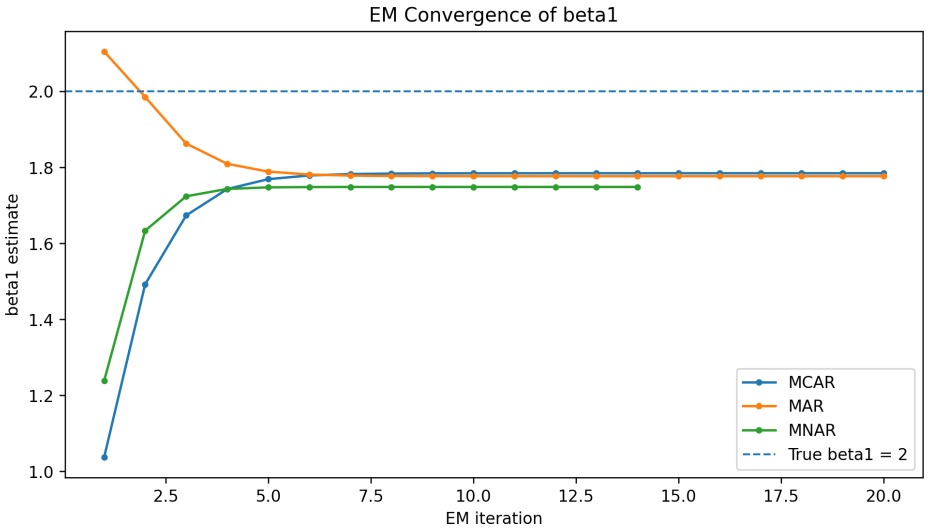


Figure 6: EM convergence of β_1 in the three missingness settings.

References

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